Bloomberg Sketchbook

5 classes · Bloomberg · Fall 2026 finals

Each class: the idea in one line, the sketch step by step, plain explanation, formulas, a worked example, common traps and where to read more.

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1. Bloomberg: passing the four tests (Bloomberg)
2. Economic indicators (Bloomberg)
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4. Fixed income and the yield curve (Bloomberg)
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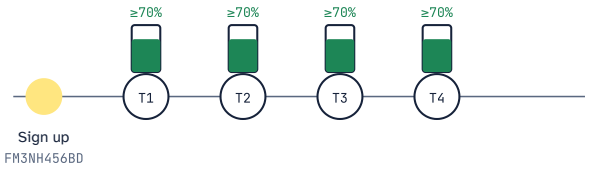
Bloomberg: passing the four tests

The Bloomberg course is pass/fail on four tests: at least 70% on each one, done by December 14, under class code FM3NH456BD.

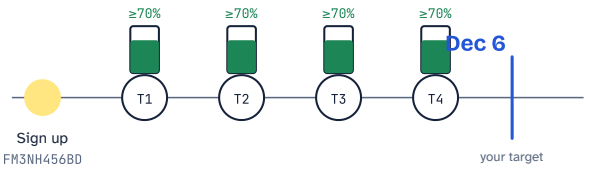
The sketch, step by step



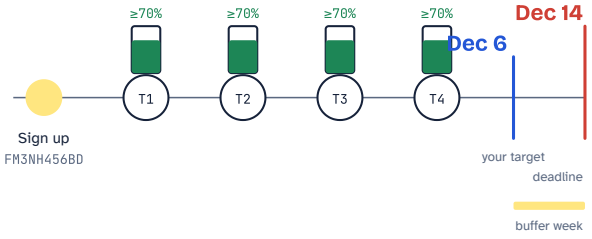
Step 1. Step 1: sign up and enter class code FM3NH456BD.



Step 2. Then four tests along the way. Each needs at least 70%.



Step 3. Your own target finish line is December 6.



Step 4. The real deadline is December 14. The week in between is your safety buffer.

What it means

- Sign up with the access guide in your folder and enter class code `FM3NH456BD` so the instructor can see your scores.
- Watching videos or getting a certificate is **not** enough. All four test scores must be 70% or more.
- Do the lessons actively: pause, note unfamiliar terms, and link them to what you learned here (yields and duration, futures, options, credit).
- Aim to finish by your own target of **December 6**, leaving a week of buffer before the real deadline of **December 14**.
- After each test, check the score shows on your class record.

The formulas

```
Pass = all four tests ≥ 70%
Deadline: Mon Dec 14, 2026 (target Dec 6)
```

Bloomberg lessons and tests are online, not in your folder.

Worked example

Question. Plan: you have 30-minute Bloomberg slots twice a week from Nov 2. How do you fit four tests before Dec 6?

1. Week of Nov 2: register, confirm the class code, start the first module.

2. Weeks of Nov 9 and Nov 16: finish modules and take tests 1 and 2.
3. Weeks of Nov 23 and Nov 30: tests 3 and 4 (book a longer session if a test needs it).
4. Retake any test under 70%, then check all four scores appear.

Answer: Four tests done by Dec 6, one week of buffer

Common traps

- Not entering the class code, so the instructor never sees your scores.
- Assuming a certificate means you passed.
- Leaving all four tests to the final week.

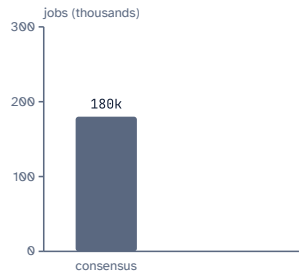
Where to read more

- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_01_Bloomberg_Signup_and_BMC_Access_Summer_2025.pdf`
- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_02_Requirements.md`

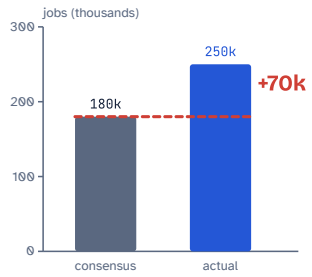
Economic indicators

Markets move on surprises, not on the news itself: what matters is the actual number minus what economists expected.

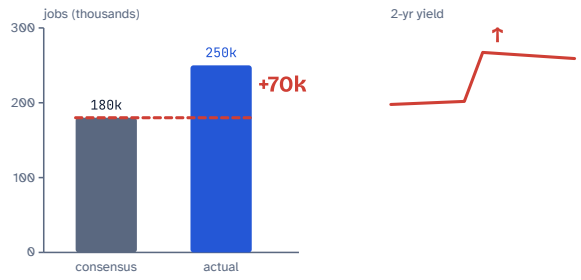
The sketch, step by step



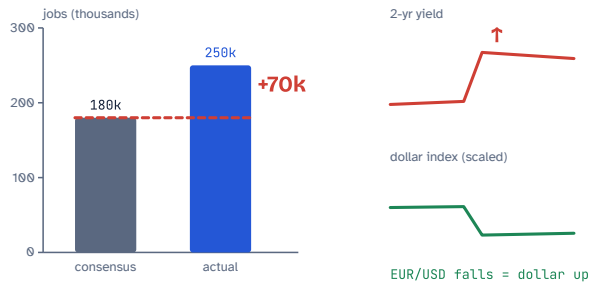
Step 1. The economists' consensus forecast: +180k jobs.



Step 2. The release: +250k. A +70k surprise.



Step 3. The market reacts within seconds: 2-year yield jumps up.



Step 4. And the dollar rises against other currencies.

What it means

- **GDP:** total output. Growth rate is the headline. Released quarterly.
- **Inflation (CPI, PCE):** how fast prices rise. Central banks target about 2%.
- **Jobs (nonfarm payrolls, unemployment rate):** monthly; one of the biggest market movers in the US.
- **Surveys (PMI, consumer confidence):** early signals. PMI above 50 = expanding.
- **Leading vs lagging:** leading indicators (PMI, building permits) turn before the economy; lagging ones (unemployment) after.
- **Central bank link:** strong growth or high inflation → expect higher policy rates → bond yields up, currency up, often stocks down.
- On the terminal, the economic calendar shows the release time, the consensus (survey) and the actual.

The formulas

```
Surprise = Actual - Consensus
Positive surprise (strong data) → yields ↑, currency ↑
Negative surprise → yields ↓, currency ↓
```

Typical reactions. Real markets can react differently if the details disagree with the headline.

Worked example

Question. Consensus for US payrolls is +180k. The actual is +250k. Unemployment is unchanged. What do you expect for Treasury yields and the dollar?

1. Surprise = 250k − 180k = `+70k`, a strong beat
2. Stronger jobs → the Fed is less likely to cut rates
3. Short-term Treasury yields `rise` (prices fall)
4. The dollar `strengthens`, as higher US rates attract money

Answer: Yields up, dollar up

Common traps

- Reacting to the level instead of the surprise.
- Mixing up month-over-month and year-over-year inflation.
- Forgetting that a bond price falls when its yield rises.

Where to read more

- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_02_Requirements.md`
- `Bloomberg Market Concepts: Economic Indicators module`

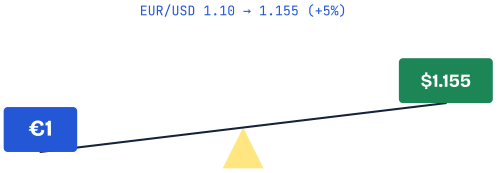
Currencies

An exchange rate is just the price of one currency in another. Interest rates, inflation, growth and risk appetite push it around.

The sketch, step by step



Step 1. A seesaw: EUR on the left, USD on the right. At 1.10, one euro balances 1.10 dollars.



Step 2. The ECB raises rates; money flows into euros. The euro side gets heavier: EUR/USD rises to 1.155.



Step 3. Cross rates: if you know EUR/USD and USD/JPY you can get EUR/JPY by multiplying.

What it means

- **Quote:** EUR/USD = 1.10 means 1 EUR (the base) = 1.10 USD (the quote currency). If it rises to 1.12, the euro got stronger.
- **Pips:** the 4th decimal (0.0001). Spread = ask − bid.
- **Interest rate differentials:** higher rates attract money → currency tends to strengthen (spot). Forwards follow interest rate parity (see the Risk class on currencies).
- **Carry trade:** borrow in a low-rate currency, invest in a high-rate one. Earns the gap, until a sudden reversal.
- **PPP (purchasing power parity):** in the long run, the same basket should cost the same everywhere; high-inflation currencies weaken.
- **Safe havens** (USD, JPY, CHF) tend to rise in a crisis.
- **Who cares:** exporters, importers, investors holding foreign assets, central banks.

The formulas

```
Amount in quote currency = amount in base × rate
Cross rate: EUR/JPY = EUR/USD × USD/JPY
% change of base = new rate / old rate - 1
```

Always write the pair as BASE/QUOTE and keep units: QUOTE per 1 BASE.

Worked example

Question. EUR/USD = 1.10 and USD/JPY = 150. (a) How many dollars is €2m? (b) What is EUR/JPY? (c) EUR/USD moves to 1.155. How much did the euro gain?

1. (a) 2,000,000 × 1.10 = `$2,200,000`
2. (b) EUR/JPY = 1.10 × 150 = `165` yen per euro
3. (c) 1.155 / 1.10 − 1 = `+5%`
4. So a US investor holding euro assets gained 5% from the currency alone.

Answer: $2.2m; EUR/JPY 165; euro +5%

Calculator example

Inputs: Amount of base currency = 2000000, Rate (quote per base) = 1.1, New rate = 1.155

```
Converted = 2,000,000 × 1.1 = 2,200,000.00
Base currency change = 1.155 / 1.1 - 1 = 5.00%
Same amount at new rate = 2,310,000.00
```

Common traps

- Multiplying when you should divide (check units: USD per EUR × EUR = USD).
- Reading a fall in EUR/USD as the dollar weakening (it is the dollar strengthening).
- Forgetting the currency effect on foreign investments.

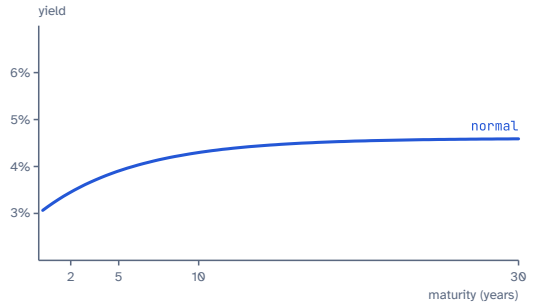
Where to read more

- Bloomberg Market Concepts: Currencies module

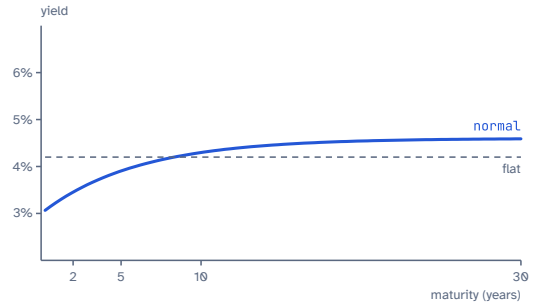
Fixed income and the yield curve

A bond's price and yield move in opposite directions. The yield curve (yields by maturity) shows what the market expects for rates and growth.

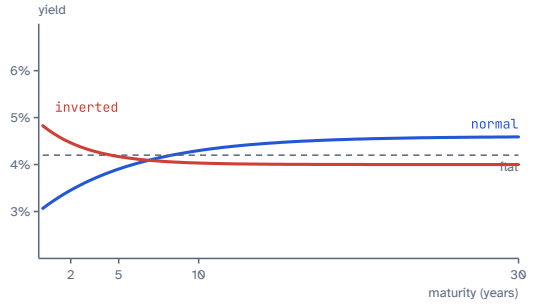
The sketch, step by step



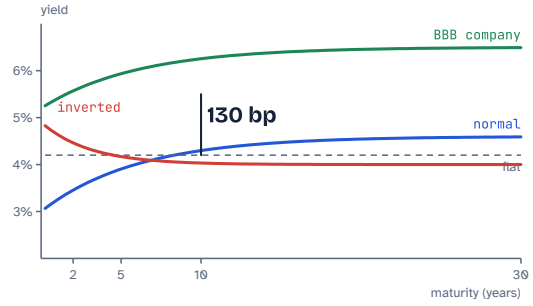
Step 1. A normal yield curve: you get paid more to lend for longer.



Step 2. A flat curve: all maturities pay about the same.



Step 3. An inverted curve: short rates above long rates. Markets expect cuts, often a recession warning.



Step 4. Add a company's curve above the Treasury curve. The gap is the credit spread: 130 bp at 10 years.

What it means

- **Price vs yield:** rates up → existing bonds with lower coupons are worth less → price down.
- **Yield curve shapes:** normal (upward: longer = higher yield), flat, inverted (short rates above long, often seen before recessions).
- **Central banks** set short rates; long rates reflect expected future short rates plus a term premium.
- **Credit spread:** corporate bond yield − Treasury yield of the same maturity. Wider spread = more perceived default risk.
- **Ratings:** AAA … BBB− is investment grade; BB+ and below is high yield ("junk").
- **Duration** (see the Derivatives class) tells you price sensitivity: duration 7 → about −7% for +1% in yield.

The formulas

```
Credit spread = corporate yield - Treasury yield
Curve slope   = 10-yr yield - 2-yr yield
% price change ≈ -Duration × Δyield
```

1 basis point (bp) = 0.01%.

Worked example

Question. 10-year Treasury yields 4.20%. A BBB company's 10-year bond yields 5.50%. The 2-year Treasury yields 4.60%. Spread? Curve shape? If the corporate bond (duration 8) sees its yield rise 0.25%, price change?

1. Credit spread = 5.50 − 4.20 = `1.30% = 130 bp`
2. Slope = 4.20 − 4.60 = `-40 bp` → the curve is inverted
3. Price change ≈ −8 × 0.25% = `-2%`
4. Inverted curve: markets expect rate cuts ahead, often linked with slowing growth.

Answer: Spread 130 bp; inverted curve; price −2%

Common traps

- Saying a bond price rises when yields rise.
- Comparing yields of different maturities when computing a spread.
- Writing 130 bp as 13%.

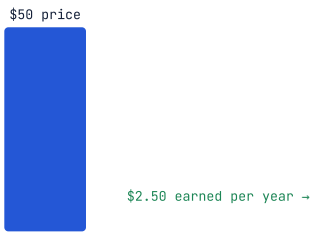
Where to read more

- `Bloomberg Market Concepts: Fixed Income module`
- `Valuation/03 Lectures/FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf`

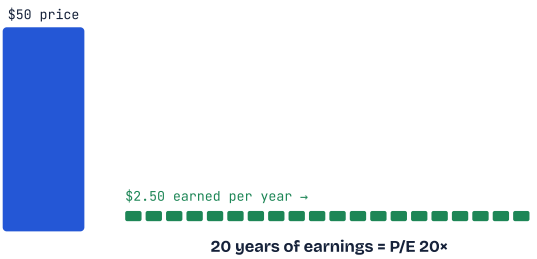
Equities and valuation multiples

A share is a slice of a company's future profits. Multiples like P/E are a quick way to compare how "expensive" companies are.

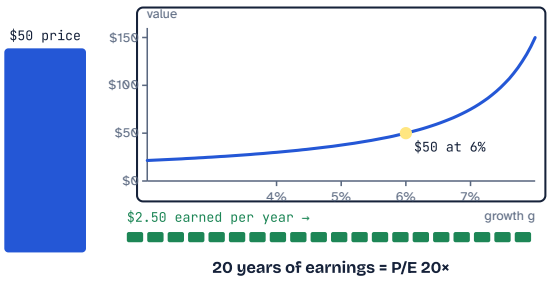
The sketch, step by step



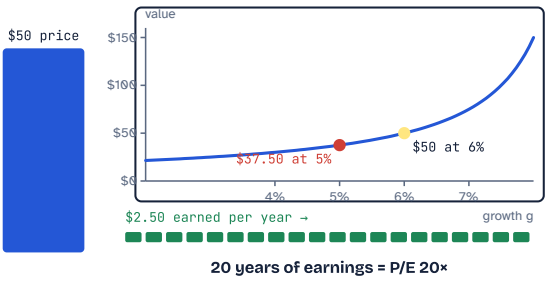
Step 1. A $50 share. Each year it earns $2.50 per share.



Step 2. So you pay 20 years of today's profit: P/E = 20×.



Step 3. The Gordon model values the growing stream of dividends: 1.50 / (9% − 6%) = $50.



Step 4. Small changes in growth move the value a lot: at 5% growth the value drops to $37.50.

What it means

- **P/E = price / earnings per share.** How many dollars you pay for $1 of yearly profit. Higher can mean faster expected growth, or overpricing.
- **EV/EBITDA:** enterprise value (equity + debt − cash) over operating earnings. Compares firms with different debt levels.
- **Dividend yield** = dividend / price. **Market cap** = price × shares.
- **DCF (discounted cash flow):** the "proper" value: PV of future free cash flows (Valuation classes). A simple version is the Gordon model: $P = D_1/(r - g)$.
- **Indices** (S&P 500, etc.) track groups of stocks. **Beta** measures how much a stock moves with the index (Portfolio/CAPM class).
- **What moves stocks:** earnings surprises, guidance, interest rates (higher rates = lower PV of future profits), sector news.

The formulas

```
P/E = Price / EPS
EV = Market cap + Debt - Cash
Gordon growth: P0 = D1 / (r - g)
```

EPS = earnings per share, D1 = next year's dividend, r = required return, g = growth rate.

Worked example

Question. A stock is $50 with EPS $2.50 and next year's dividend $1.50. Required return 9%, dividends grow 6% a year. P/E? Gordon value? Is it cheap?

1. P/E = 50 / 2.50 = `20×`
2. Gordon value = 1.50 / (0.09 − 0.06) = 1.50 / 0.03 = `$50`
3. Model value = market price, so fairly priced under these assumptions.
4. If growth were only 5%: 1.50 / 0.04 = $37.50, so the price assumes 6% growth.

Answer: P/E 20×; Gordon value $50 (fairly priced)

Calculator example

Inputs: Price = 50, EPS = 2.5, Next dividend D1 = 1.5, Required return (%) = 9, Growth (%) = 6

```
P/E = 20.0×
Gordon value = 1.5 / (0.09 - 0.06) = $50.00
```

Common traps

- Comparing P/E across very different industries.
- Using Gordon when g ≥ r (the formula breaks).
- Forgetting to subtract cash and add debt for EV.

Where to read more

- `Bloomberg Market Concepts: Equities module`
- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`

Watch and learn online

Free videos and courses to use alongside these classes. Coursera courses are free when you choose "audit".

Bloomberg

- **Bloomberg for Education portal (Bloomberg Market Concepts)** (The actual course and tests)
  https://portal.bloombergforeducation.com/
  Sign in here, enter class code FM3NH456BD, and do the modules. The core BMC modules are Economic Indicators, Currencies, Fixed Income and Equities; check the course instructions to confirm these are your four required tests.
- **WallStreetMojo: Bloomberg Market Concepts beginner's guide** (Free written guide)
  https://wallstreetmojo.com/bloomberg-market-concepts-bmc
  Explains what each BMC module covers and how the tests work, before you start.
- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Helps with the bond, stock and currency basics the BMC modules assume.
- **Yale (Coursera): Financial Markets with Robert Shiller** (Free to audit, video course)
  https://www.coursera.org/learn/financial-markets-global
  Background on how stock, bond and currency markets work and react to the economy.